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**George Washington University in Partnership with Ernst & Young US LLP**

**DNSC 6317 - Business Analytics Practicum**

## **AI-Driven Fairness Measurement and Summarization in Large Language Models**

### **An Examination for Financial Institutions**

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## Executive Summary

As financial institutions adopt large language models (LLMs) to assist with internal tasks including summarizing client data for further processing, concerns around fairness and bias in AI-generated (Artificial Intelligence generated) outputs are growing. In recent years, generative AI tools such as LLMs have been integrated across industries, offering powerful capabilities for automating tasks like summarization, sentiment detection, and information extraction. Concerns about potential systemic biases embedded within these models have become increasingly prominent in high-stakes domains such as financial services, where ethical, legal, and regulatory standards demand fairness and transparency.

This project presents a structured and reproducible framework for measuring and mitigating potential bias in LLM-generated outputs. Using financial fraud-related stories from the CNN/DailyMail dataset, we employed a substitution-based method to simulate identity variations across different protected demographic classes. Summaries were generated via the Cohere Command R model hosted on AWS Bedrock, and were quantitatively evaluated using statistical metrics such as Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Bidirectional Encoder Representations from Transformers Score (BERTScore) and Adverse Impact Ratio (AIR).

Statistical techniques, including Multivariate Analysis of Variance (MANOVA), Analysis of Variance (ANOVA), and Canonical Correlation Analysis (CCA), revealed measurable differences in summary outputs, particularly for white female identities. Sentiment analysis and human evaluation confirmed that these disparities are not merely statistical artifacts but are reflected in the tone and framing of generated summaries. Despite overall AIR scores suggesting no systemic disadvantage, localized inconsistencies highlight the importance of further testing.

To that end, the report proposes a further multi-phase testing roadmap including model testing, red teaming, and field testing along with recommendations to adopt Human-Calibrated Automated Testing (HCAT) frameworks. This playbook provides financial institutions with a scalable, modular approach to audit LLM outputs for fairness, supporting responsible AI deployment in line with regulatory expectations.

## Problem Understanding

LLMs are being adopted rapidly across financial institutions for tasks involving large-scale unstructured data, including customer communication, compliance reviews, and fraud detection. However, without rigorous fairness assessments, these models risk reinforcing systemic biases, particularly those related to race and gender potentially leading to unethical or even illegal decision-making processes. To address this issue, Earnst & Young (EY) is seeking a comprehensive playbook, including a set of techniques for defining fairness outcomes as a function of the use case and associated metrics for fairness measurement, that could be deployed by their clients to risk-assess LLM for novel use cases

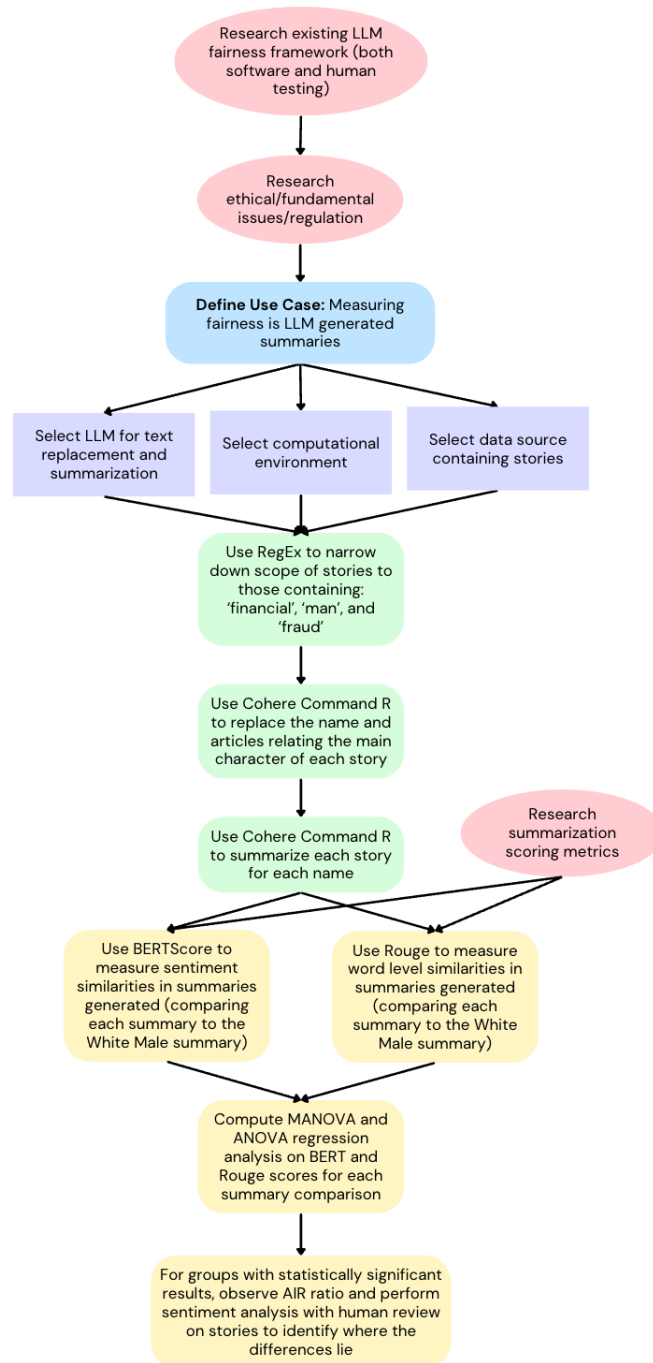
The central problem addressed in this paper is the lack of a standardized, scalable methodology to evaluate and quantify bias in LLM-generated outputs, especially in sensitive use cases within the financial sector. Despite advancements in model performance, institutions lack actionable tools to audit and mitigate identity-based disparities in AI-generated narratives. This report investigates whether LLMs generate meaningfully different summaries when only the identity (race or gender) of a central character is changed despite all other content remaining constant. This nuanced problem demands both linguistic sensitivity and statistical rigor, as the consequences of biased outputs may affect public trust, regulatory compliance, and real-world outcomes. The solution must be reproducible, adaptable across platforms and models, and aligned with legal fairness standards such as those outlined in the Equal Credit Opportunity Act. Ultimately, this work seeks to empower stakeholders in the financial industry with practical tools to detect, understand, and address potential biases embedded in generative AI systems.

## Methodology

For this project, we focused on the use case to be specifically for the financial services industry, utilizing AWS Bedrock and analytical techniques including ANOVA, MANOVA, AIR, and Sentiment Analysis to develop a comprehensive framework. The concept map in **Figure 1**, outlines the steps taken to reach our final playbook. Through rigorous research, we identified the optimal LLM, cloud computing environment, and evaluation metrics to ensure adaptability across various applications ultimately creating

an eight-step fairness and bias testing playbook. The following methodology sections provide a detailed walkthrough of these eight steps in order.

**Figure 1: Concept Map**



*Note. The concept map outlines the eight step approach to our ethical and fairness testing playbook.*

## Software Selection

Our key selection criteria for the language model prioritized an open-source LLM capable of summarizing large quantities of unstructured data in the form of stories, that can be scaled to a higher computational environment. Based on this criteria, we decided on one primary LLM: Cohere Command R. The Cohere Command family of models is instruction-following conversational models that generate text based on user instruction. The LLMs have abilities involving retrieval augmented generation (RAG), multilingual functionality, chatbot-like functionality, and tool use. The Cohere Command R model specifically is proficient at text summarization and can accept upwards of 128k context length, providing a maximum of 4k output tokens (*The Command R Model (Details and Application)* — Cohere, 2024), making it an ideal model for our project.

Since the primary objective for this project is to utilize the LLM for summary generation rather than forecasting or predictive analysis, having a model that specializes in financial fine-tuning was not a critical requirement. Given our use case also involves instruction based name and article substitutions for various demographic groups, the summarization and instruction based functionalities of Cohere Command R made it the most appealing. Cohere Command R is a strong general model that can be used on a wide range of text inputs.

Initially, we experimented with using LLaMA 3.1 as the primary LLM for this project; however, it did not consistently generate accurate or concise summaries. Despite providing a simple and clear prompt such as, “generate a 3 sentence summary,” LLaMA frequently produced outputs exceeding the sentence limit or simply restated the original story without meaningful abstraction. Due to these limitations in controllability and summarization quality, we opted to move forward with Cohere, which demonstrated superior performance in producing coherent, concise, and contextually appropriate summaries. As a result, Cohere was selected as the final LLM to be tested for this project.

In parallel with our model research and selection, we evaluated computational environments capable of automating and executing our playbook, as its resource requirements would exceed the capacity of personal machines. After conducting thorough research on AWS and guaranteed access to its

resources through our university affiliation, AWS emerged as the most practical choice. Additionally, AWS is a widely adopted industry-standard cloud computing platform, making it highly relevant to EY's clients.

AWS Bedrock is a cloud service within AWS that simplifies scaling generative AI applications. It supports high-performing foundation models from providers like Cohere, Anthropic, DeepSeek, and Meta (*Build Generative AI Applications with Foundation Models - Amazon Bedrock - AWS*, 2023). This ensures the playbook remains adaptable to different LLMs based on specific business needs. While AWS is the preferred environment for development and testing, our playbook is designed to allow for easy transfer to alternative cloud platforms as needed.

## Data Analyzed

The objective of this playbook is to evaluate risks associated with LLMs in novel use cases while ensuring flexibility, scalability, and alignment with ethical principles. We elected to focus on unstructured data, rather than structured data, as it more accurately reflects the types of complex, free-text inputs commonly encountered in the financial services industry, such as client communications, transaction notes, and compliance reports, where critical insights often lie beyond traditional tabular formats. To initiate this process, we utilized a publicly available dataset from Hugging Face, *The CNN/DailyMail Dataset*, which contains over 300,000 news articles authored by journalists from *CNN* and the *Daily Mail* (See, 2025). This dataset was deemed well-suited for our analysis due to its applicability to both abstractive and extractive summarization tasks.

To demonstrate our proof of concept, we utilized regular expressions (RegEx) in Python, a powerful tool used to search, match, and manipulate text patterns using the “re” module to identify relevant news articles within the training dataset, specifically containing the keywords “Man,” “Fraud,” and “Financial” to ensure that the selected stories featured a male character involved with fraudulent activities in a financial setting. This approach also refined our scope from general CNN articles to those specifically related to fraudulent financial activity and captured stories with a clear male role that could be easily modified. This filtering step was essential in developing a reproducible framework, ensuring its applicability to new datasets of unstructured financial data.

After retrieving a dataset of 100 CNN news stories for analysis using the predefined specifications, we processed these narratives through the Cohere LLMs, to modify the main character's identity to reflect the protected classes of gender and three races. To establish a baseline for comparison, we prompted Chat Generative Pre-trained Transformer (ChatGPT) to identify the most common full name of a white male in the United States, selecting "James Johnson" as our primary case (OpenAI, 2025). Similarly, we obtained the most common and "traditional" name for a second white male to be David Miller. The comparison between the two male names serves as the control group. To account for protected classes, we prompted ChatGPT to identify a common name for each class that is distinct from typical white male names. Based on this approach, we selected the following names as a proxy for each race observed in this report: Mary Smith for a white female, Jamal Washington for a Black male, Wei Chen for an Asian male, and Jose Rodriguez for a Hispanic male. We further prompted the LLM to adjust pronouns accordingly to match each identity. This procedure resulted in six different sets of otherwise identical stories, differing only in the identity of the main character and their pronouns. Each story was then processed through the Cohere LLM, prompting the model to generate a three-sentence summary. These summaries served as the foundation for our subsequent comparative analysis.

As outlined in Fair Lending Laws, such as the Equal Credit Opportunity Act and the Fair Housing Act, discrimination based on race, national origin, religion, sex, and other protected classes is strictly prohibited (Federal Deposit Insurance Corporation, 2024). In this playbook, we test only gender and race, however this can be easily expanded to cover other protected classes identified in Fair Lending Laws.

Building on these principles and insights from a recent Cornell study, we applied a name substitution method to simulate variation across protected classes, based on findings that LLMs respond differently to names that sound demographically distinct (Nyarko, 2024), a phenomenon that also applies to genders. One study found that LLM responses systemically disadvantage names that are commonly associated with racial minorities and women (Haim et al., 2024). Another study by the National Bureau of Economic Research altered the perceived race of job applicants by replacing names on resumes to either distinctly white or distinctly African American-sounding names. The results revealed that even subtle name changes led to significant differences in treatment (Bertrand & Mullainathan, 2003). Thus, to assess



potential racial and gender bias in LLMs, we prompt the model to replace a main character's name with one that suggests a different demographic identity.

### **Analytics Techniques Used**

Since the aim of this playbook is to develop a framework that quantifies the difference in text generated for an individual in the control group versus a protected group, we propose using two key scoring metrics: ROUGE and BERTScore. These metrics will be used to compare the summaries produced for a protected group against those of the control group, allowing us to calculate and assess any differences.

ROUGE is an evaluation metric for generated text which measures the overlap between the generated summary and a reference summary based on n-grams, providing scores for precision, recall, and F1-score. The metric is computed by calculating the ratio of occurrences in both phrases to the occurrences in the reference phrase only, thereby capturing the overlap between the two. There are three ROUGE metrics, each consisting of a unique calculation to measure precision, recall, and F1-score. Among these, the ROUGE-1 metric focuses on similarities between the individual words (unigrams) in the phrases and the ROUGE-2 metric introduces a unique component by counting similarities between consecutive word pairs (bigrams) and comparing across the two phrases. Lastly, the ROUGE-L metric compares the largest common subsequence of words in the two phrases (Chiusano, 2022). We focus on F1-scores for each of these ROUGE variants as they represent the harmonic mean of precision and recall, providing a balanced view of how much relevant content is captured (recall) and how much irrelevant content is avoided (precision). This provides additional benefit by considering more of the overall sentence structure as opposed to looking at individual words or pairs of words.

BERTScore is another automated metric that computes text similarity. The model calculates the similarity of phrases as a sum of cosine similarities between token embeddings (Zhang et al., 2020). A large advantage of BERTScore lies in its ability to identify similarities based on BERT contextual embeddings. This means that even if exact words are not included in each phrase, words with similar meaning will be identified as similar. Unlike ROUGE, which is based on surface-level token overlap,

BERTScore captures deeper, contextual understanding of meaning. For each comparison, we calculate precision, which measures how much of the generated content is semantically aligned with the reference, recall, which measures how much of the reference content is captured in the generated text, and F1-score, which is a balanced measure of precision and recall. For our use case, we use the DeBERTa (Decoding-enhanced BERT with disentangled attention) model instead of RoBERTa (Robustly Optimized BERT Approach) or other BERT variants due to its superior performance in capturing nuanced semantic relationships. DeBERTa introduces disentangled attention and enhanced position modeling, which are particularly beneficial for comparing subtle differences in summary phrasing across demographic variations (Vyacheslav Efimov, 2023).

For each demographic comparison group (e.g., white female, Black male, Asian male), 100 summaries were generated and compared against 100 reference summaries based on a white male character, James Johnson. Each summary pair yields six scores - 3 ROUGE F1 scores (ROUGE-1, ROUGE-2, ROUGE-L) and 3 BERTScore metrics (precision, recall, F1). This results in 600 rows of evaluation data per demographic group, forming a consolidated dataframe of 3,000 scores across five groups (excluding the original reference group). The computed ROUGE and BERTscores serve as a quantitative basis for downstream statistical analysis. By comparing score distributions across demographic groups, we can detect whether the LLM-generated summaries differ significantly in linguistic structure or semantic content. We used the computed ROUGE and BERTscores as inputs for statistical analysis, beginning with correlation analysis to examine relationships among the metrics, followed by MANOVA to detect multivariate differences across protected classes. Based on MANOVA results, we conducted individual ANOVAs to identify which specific scores contribute most to observed disparities. Summaries from groups that show statistically significant differences in these analyses were then flagged for deeper examination through sentiment analysis and manual review, allowing us to assess whether linguistic tone or sentiment shifts may contribute to potential bias in the model's outputs.

The next part of our analysis computes the average AIR for each metric across the different classes observed in this dataset. In its classical application, AIR is a metric that compares the rate of favorable outcomes for a protected class compared to those for a control group as shown in **Equation 1**

(Insurance Circular Letter No. 7 RE: Use of Artificial Intelligence Systems and External Consumer Data and Information Sources in Insurance Underwriting and Pricing, 2024). In this playbook, we calculate AIR as shown in equation 1 below. The scores used include BERTScore and ROUGE, which assess the similarity of generated summaries when comparing each class to the base case, white\_male1. To establish a reference score, we compare the summaries generated for white\_male1 and white\_male2. These reference scores represent the expected level of overlap when prompting the LLM to produce summaries. Thus, the values used to compute AIR ultimately compare the rate of favorable outcomes for each protected class relative to the reference group, with this idea of a favorable outcome identified as the degree of overlap between a class's summary and that of white\_male1.

**Equation 1: AIR Computation**

$$AIR = \frac{\text{Score for Protected Class}}{\text{Score for Reference Group}}$$

*Note.* The AIR is a metric that compares the rate of favorable outcomes for a protected class compared to those for a control group.

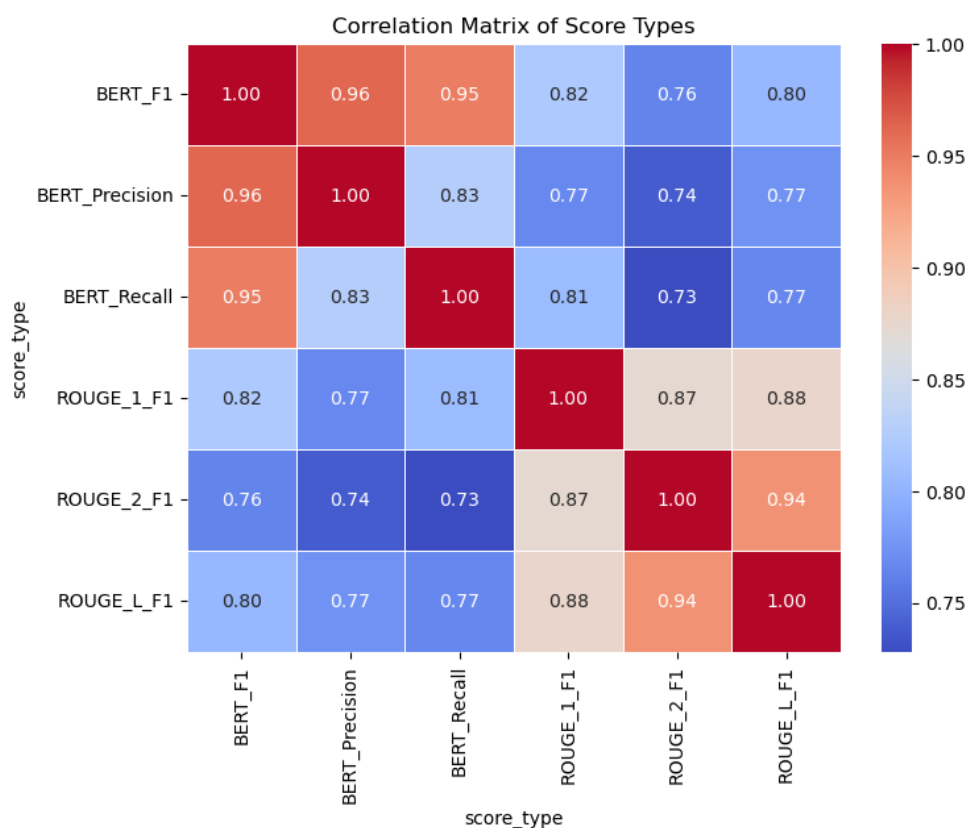
Here, we choose to compute AIR because it is a widely accepted and statistically validated metric with a legal precedent. According to the four-fifths rule, a selection procedure has an adverse impact if it results in a selection rate for any protected group that is less than 80% of the rate of the group with the highest selection rate. In our analysis, we set the AIR threshold to 0.90 based on trending industry standards and considered the average AIR of a metric to be satisfactory if it is above 0.9.

The final steps of the framework include sentiment analysis followed by a human review. Here, we apply TextBlob sentiment analysis to compute polarity scores for each generated summary. This provides a standardized measure of how positive or negative the language in each summary is across the protected group and control group. Finally, a manual review of the summaries that exhibit the highest sentiment difference scores further confirms any difference in tone and meaning between 2 classes to identify bias.

## Results and Conclusions

As mentioned previously, the evaluation began with a correlation analysis (correlation matrix), revealing strong internal consistency among metrics as shown in **Figure 2**. BERT-based scores were highly correlated with one another (correlation between BERT F1 and BERT precision being 0.96 and correlation between BERT F1 and BERT recall being 0.95), as were ROUGE scores. Cross-metric correlations between ROUGE and BERT ranged from 0.76 to 0.82, suggesting that while the metrics are related, each captures distinct features of semantic and lexical similarity, justifying their inclusion in a combined analysis.

**Figure 2:** Correlation Matrix Between ROUGE and BERTScore Types



*Note. The heatmap displays the Pearson correlation coefficients among BERT-based and ROUGE-based F1 scores. Darker red colors represent higher positive correlations, while darker blue colors indicate lower correlations.*

To test for group-level differences, we employed a MANOVA model using BERT F1, ROUGE-1 F1, ROUGE-2 F1, and ROUGE-L F1 as input variables. BERT precision and recall were excluded to avoid redundancy, as BERT F1 is a harmonic balance of the two. The MANOVA, using Roy's Greatest Root, yielded a p-value of 0.0076 as shown in **Table 1**, indicating a statistically significant difference in the linear combination of scores across at least one protected class at the  $\alpha = 0.05$  threshold and implies at least one protected class exhibits statistical significance for different scores. This justified further exploration via individual ANOVAs to determine which score is driving the statistical significance or whether it is the combination of scores that is capturing a unique difference.

**Table 1: MANOVA Results**

| Statistics             | Value  | Num DF | Den DF | F Value | Pr > F |
|------------------------|--------|--------|--------|---------|--------|
| Wilks' lambda          | 0.9541 | 16     | 1504   | 1.4552  | 0.1082 |
| Pillai's trace         | 0.0465 | 16     | 1980   | 1.4542  | 0.1081 |
| Hotelling-Lawley trace | 0.0474 | 16     | 978    | 1.4555  | 0.1090 |
| Roy's greatest root    | 0.0285 | 4      | 495    | 3.5223  | 0.0076 |

*Note. MANOVA results show Roy's greatest root as statistically significant with a p-value of 0.0076.*

Subsequent one-way ANOVA models revealed that neither BERT F1 nor ROUGE-1 F1 or ROUGE-L F1 showed statistically significant differences between the white\_male2 (David Miller, reference group) and any other demographic group ( $p > 0.05$ ). However, displayed in **Table 2** below, ROUGE-1 F1 exhibited a significant difference ( $p = 0.023$ ) between the white\_male2 and white\_female group, suggesting that summaries for this group diverged in unigram-level word choices compared to the baseline and indicating potential bias in the summaries associated with that demographic.

**Table 2: ROUGE-1 F1 ANOVA Results**

| Model Summary - OLS Regression Results |                  |                    |        |
|----------------------------------------|------------------|--------------------|--------|
| Dep. Variable                          | ROUGE_1_F1       | R-squared          | 0.021  |
| Model                                  | OLS              | Adj. R-squared     | 0.013  |
| Method                                 | Least Squares    | F-statistic        | 2.654  |
| Date                                   | Tue, 08 Apr 2025 | Prob (F-statistic) | 0.0325 |
| Time                                   | 23:58:33         | Log-Likelihood     | 489.46 |

|                  |           |     |        |
|------------------|-----------|-----|--------|
| No. Observations | 500       | AIC | -968.9 |
| Df Residuals     | 495       | BIC | -947.8 |
| Df Model         | 4         |     |        |
| Covariance Type  | nonrobust |     |        |

| Protected Class         | coef    | std err | t      | P> t  | [0.025 | 0.975] |
|-------------------------|---------|---------|--------|-------|--------|--------|
| Intercept – white_male2 | 0.4231  | 0.009   | 46.307 | 0     | 0.405  | 0.441  |
| asian_male              | -0.0009 | 0.013   | -0.067 | 0.947 | -0.026 | 0.025  |
| black_male              | -0.0094 | 0.013   | -0.728 | 0.467 | -0.035 | 0.016  |
| hispanic_male           | 0.0101  | 0.013   | 0.785  | 0.433 | -0.015 | 0.036  |
| white_female            | -0.0295 | 0.013   | -2.282 | 0.023 | -0.055 | -0.004 |

*Note. ROUGE-1 F1 ANOVA results show a statistically significant difference in average ROUGE-1 F1 values between white\_male2 and white\_female with a p-value of 0.023.*

Given the complexity of inter-metric relationships, we then conducted CCA to better understand how combinations of evaluation scores relate to protected class labels. CCA identifies the relationship between the scores and protected classes through a weighted linear combination. It reveals which groups of scores are most strongly correlated with protected class, helping to reduce dimensionality by eliminating scores with the weakest correlations. The overall canonical correlation value of 0.1664, as shown in **Table 3** below, suggests a weak to moderate relationship between the evaluation metrics and protected class.

**Table 3: CCA Coefficients and Results**

|                                            |        |        |        |        |
|--------------------------------------------|--------|--------|--------|--------|
| <b>Canonical Correlation Coefficients:</b> | 0.1664 | 0.1095 | 0.0770 | 0.0295 |
| <b>Largest Canonical Correlation:</b>      | 0.1664 |        |        |        |

*Note. The overall canonical correlation value of 0.1664 suggests a weak to moderate relationship between the evaluation metrics and protected class.*

Furthermore, ROUGE-L F1 had the weakest canonical weight (0.0295) and was thus excluded from the final MANOVA model we ran. Using the CCA-derived weights, we reweighted the original scores and updated the analysis as shown in **Table 4**. The revised MANOVA again demonstrated statistical significance, with the p-value associated with Roy's Greatest Root (0.0076) remaining below 0.05, reinforcing the presence of measurable disparities in how the LLM summarizes content for different demographic groups.

**Table 4:** Weighted MANOVA Results using ROUGE-1 F1, ROUGE-2 F1 and BERT F1

| Statistics             | Value  | Num DF | Den DF | F Value | Pr > F |
|------------------------|--------|--------|--------|---------|--------|
| Wilks' lambda          | 0.9572 | 12     | 1305   | 1.8117  | 0.0417 |
| Pillai's trace         | 0.0433 | 12     | 1485   | 1.8107  | 0.0417 |
| Hotelling-Lawley trace | 0.0442 | 12     | 858    | 1.8125  | 0.0422 |
| Roy's greatest root    | 0.0284 | 4      | 495    | 3.5186  | 0.0076 |

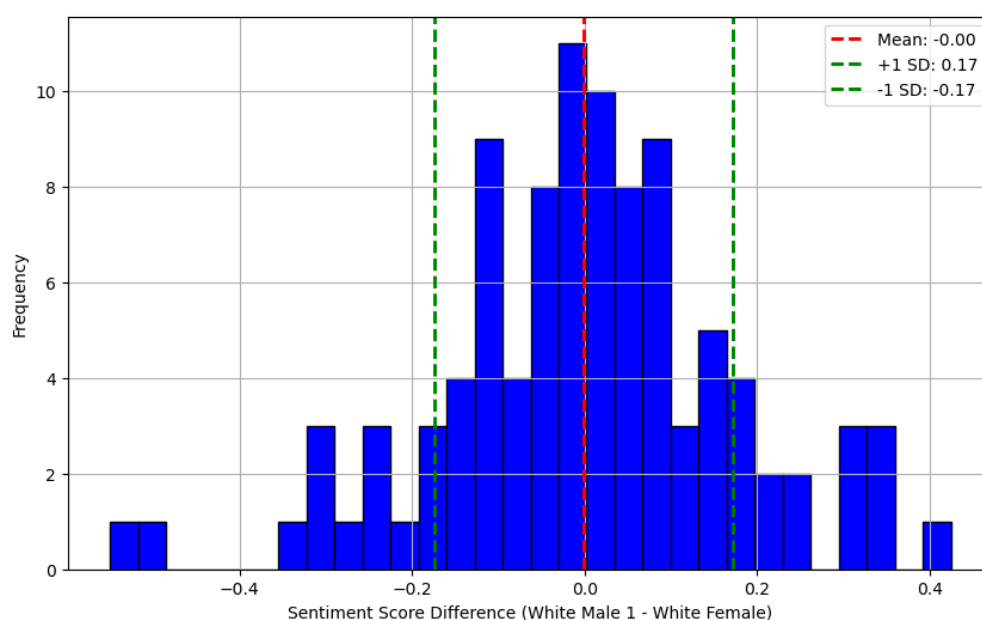
*Note. The weighted MANOVA results demonstrated statistical significance, with the p-value associated with Roy's Greatest Root (0.0076).*

As step 6 of our analysis, we computed the average AIR values across all protected classes to test for any disproportionate impact on any class. The AIR value of interest is the ROUGE-1 F1 for the white\_female group, since this combination of score and group demonstrated statistical significance in the ANOVA results. As can be seen in **Appendix H**, the results showed that the average AIR values across each group observed were found to be above 0.90. This indicates that, from a fairness auditing perspective, there is no strong evidence of systemic or disproportionate disadvantage for any group in this particular dataset using Cohere as the LLM. The AIR findings provide reassurance that, in aggregate, the LLM outputs are reasonably balanced.

Based on the statistical analysis of individual ROUGE and BERT metrics and the subsequent AIR ratio computations, while we do not detect bias, we cannot reasonably conclude that the entire dataset is free of bias. The statistically significant difference observed in ROUGE-1 F1 scores for the white\_female group, along with metric-specific disparities identified through ANOVA and sentiment analysis, point to localized inconsistencies in the model's behavior.

However, these findings indicate that subtle, identity-linked variations in language and framing do exist and warranted further investigation, prompting us to run a sentiment analysis followed by a human evaluation of summaries, particularly where statistical signals indicate potential bias in model behavior. To do this, we applied sentiment analysis using TextBlob to each generated summary. This tool provides a sentiment polarity score ranging from -1 (most negative) to +1 (most positive) (ES, 2023). We computed sentiment scores for each summary across all demographic groups and then calculated the sentiment difference between each protected class and the reference group (white\_male2). A histogram of the sentiment score differences between white\_male1 and white\_female are shown in **Figure 3** below.

**Figure 3:** *Distribution of Sentiment Score Differences Between white\_male1 and white\_female*



*Note. The distribution shows the sentiment score differences (white\_male1 – white\_female). The red dashed line indicates the mean difference (approximately 0.00). Green dashed lines represent  $\pm 1$  standard deviation from the mean ( $\pm 0.17$ ). Negative values indicate lower sentiment scores for white\_male1 compared to white\_female.*

As a next step, we identified the top 10 summaries with the most positive and most negative sentiment score differences since we previously identified statistical significance in the ROUGE-1 F1 scores of the white\_female group shown in **Appendix I** below. These outliers were manually reviewed to



assess whether the tone, framing, or emotional content of the summaries differs systematically across demographic identities. Upon the human review of the 20 identified summaries, we observed that among the top 10 summaries with the highest negative sentiment score differences, there was a clear difference in tone in how the model summarized the stories for a male versus a female protagonist. For example, for summary IDs 73 and 99, we observed that the model considered the white\_male1, James Johnson, as the person accused of committing the crime when summarizing the story with a male protagonist but considered the white\_female, Mary Smith, as the victim of the crime when summarizing the story with the female protagonist. This qualitative review helps validate whether the observed statistical and structural differences reflect substantive bias in how individuals are portrayed. A complete set of the remaining statistical results is available in **Appendices A–G**. While these results did not have a direct impact on our selected dataset, they may prove valuable when applying this playbook to new data and tested via a different LLM.

### Potential Next Steps

Frameworks for testing bias in LLMs can be categorized into three types - model testing, field testing, and red teaming. This paper focuses heavily on model testing by analyzing BERTscores and ROUGE scores of various summaries generated by the LLM, Cohere Command R in this case, and interpreting the results in real world scenarios. However, field testing and red teaming are important frameworks that further solidify the analysis of LLM outputs.

### Red Teaming

Following sentiment analysis and human evaluation of flagged summaries, a valuable next step is to conduct red teaming, a structured adversarial testing process designed to uncover hidden biases, edge cases, and failure modes in LLM-generated outputs. By intentionally probing the model with carefully crafted inputs across diverse demographic contexts, red teaming can help stress-test the system's robustness, reveal subtler forms of bias, and provide actionable insights for model refinement and safer deployment. Red teaming involves adversarial interactions with AI applications to test their resilience against potential risks and violations of safety guardrails (Vongthongsri, 2025). Unlike conventional model

testing, which relies on structured prompts and automated evaluations, red teaming introduces human creativity to probe AI vulnerabilities by intentionally crafting complex, misleading, or ethically sensitive scenarios. The process includes dialogues between red teamers and the AI model, documentation of successful exploitation attempts, and post-session assessments to analyze potential weaknesses (Reva Schwartz et al, 2024). Red teaming provides crucial insights into how AI models may respond to high-risk prompts and helps refine mitigation strategies before deployment.

In the financial services industry specifically, many institutions have developed their own frameworks to test security vulnerabilities. Threat Intelligence-Based Ethical Red Teaming Framework (TIBER-EU) was adopted by the European Central Bank's Governing Council to evaluate cybersecurity defenses in financial institutions across the EU, with a variant, TIBER-RO, implemented in Romania. Adversarial attack simulation exercises (AASE), introduced by the Association of Banks in Singapore (ABS) in 2018, is designed to rigorously challenge the cybersecurity mechanisms of financial organizations. Meanwhile, CBEST, developed by the Bank of England, employs a non-informed, covert penetration testing approach to simulate external threats against financial institutions (Top 3 Red Teaming Frameworks (TIBER,AASE,CBEST) - BreachLock, 2020). These frameworks highlight the importance of structured red teaming efforts in identifying vulnerabilities and strengthening system security, principles that can also be applied to AI model testing.

## Field Testing

Another important follow-up is field testing, where the model is deployed in a controlled, real-world setting to observe how it performs with live, unstructured data inputs and user interactions. This step allows organizations to evaluate model behavior in practical scenarios, monitor for unintended consequences, and gather user feedback. Field testing is essential for validating that fairness interventions from prior stages, such as reweighting or content adjustments, translate effectively into consistent, equitable performance in real-world applications. Field testing examines AI applications in real-world conditions by allowing users to interact with the system in a conventional manner while completing specific tasks. Unlike red teaming, which focuses on adversarial interactions, field testing captures natural usage patterns and user experiences. The process includes collecting dialogues, post-

session questionnaires, and annotating AI-generated outputs to understand user perceptions of model accuracy and relevance (Reva Schwartz, 2024). Field testing can include thousands of human participants interacting with AI applications in realistic settings across multiple sessions under test or control conditions. Statistical analysis, both descriptive and inferential, helps assess differences in user experiences across demographic groups and real-world contexts. Field testing is essential for identifying unanticipated biases, evaluating the effectiveness of safety guardrails, and refining AI models based on real-world interactions.

Financial institutions can conduct field testing on their AI models by deploying them in real-world banking environments where customers and employees interact with them under typical usage conditions. This involves monitoring AI-driven tools such as chatbots, fraud detection systems, and credit risk assessment models as they assist users in making financial decisions. Large-scale testing with diverse customer demographics allows institutions to gather user feedback, identify unintended biases, and evaluate the model's reliability across different financial scenarios. By analyzing transaction patterns, customer queries, and AI-generated recommendations, financial institutions can assess how well the models align with regulatory requirements, ethical guidelines, and user expectations. Statistical analyses of user interactions help fine-tune AI performance, ensuring the technology enhances financial services while maintaining fairness, security, and compliance.

### **Human vs LLM Interpretation**

One of the most promising extensions of our framework involves incorporating Human-Calibrated Automated Testing (HCAT), a structured approach for validating GLMs (generative language models) in high-stakes settings like finance. As introduced by Sudjianto et al. (2024), HCAT combines automated testing using embedding-based metrics with periodic human evaluation to align machine-generated assessments with real-world interpretations. This two-stage calibration through probability calibration and conformal prediction enables more trustworthy and explainable model validation.

Our project already explores this concept in a foundational way. By generating summaries for identical stories that differ only in the demographic identity of the main character (e.g., "James Johnson"

vs. “Mary Smith”), and analyzing variations using metrics such as ROUGE and BERTScore, we’ve implemented a basic form of automated bias detection. However, to build confidence in these findings and better simulate regulatory expectations, future iterations of the framework could incorporate human evaluators to assess whether perceived tone, sentiment, or completeness of summaries aligns with the machine-generated scores. In this context, the HCAT framework offers a structured pathway for transforming our current model testing into a human-calibrated evaluation loop, making fairness analysis not only scalable but also explainable. This step will be critical if financial institutions are to adopt generative AI responsibly and in compliance with regulatory standards.

### **Risk Considerations**

While the methodology outlined in this project offers a structured and quantitative approach to assessing fairness in LLM-generated summaries, it is important to acknowledge several risks and limitations that may affect the interpretation of our results. These risks span the entire pipeline, from data preprocessing and model prompting to evaluation metrics and multi-step generation. Addressing these limitations is crucial to ensure that our fairness assessments are both reliable and actionable, and to guide improvements in future iterations of the framework. The following sections outline key areas of potential risk and the rationale behind each.

#### **Risk of Incorrect Identity Substitution by the LLM**

One key risk in our methodology lies in the potential inconsistency of the LLM responses to demographic prompts. While prompts were designed to substitute the main character’s gender or racial identity, the LLM (Cohere, in this case) may not have reliably updated all relevant mentions in the narrative, including names, pronouns, or contextual cues. Incomplete or inconsistent identity substitution introduces noise into the analysis, making it difficult to attribute differences in summary output strictly to the demographic change. To mitigate this, we incorporated automated validation, such as keyword matching for names and pronouns, or manual spot checks as the last step of our framework to ensure the modified input stories fully reflect the intended demographic changes before summarization. While our validation efforts were piloted on a single comparison case (white\_male1 to white\_female), the same

approach can be generalized across other demographic groups to strengthen the reliability of future analyses.

### **Limitations of BERTScore and ROUGE Metrics**

Another consideration is the reliance on ROUGE and BERTScore metrics, which, while widely used in natural language processing, have inherent limitations. As mentioned previously in this report, ROUGE primarily measures lexical overlap (n-gram similarity), and BERTScore uses contextual embeddings to estimate semantic similarity. However, both can be sensitive to rephrasing, stylistic variation, or subtle differences in tone, and may not fully capture nuanced shifts in meaning or framing. Additionally, neither metric is designed to explicitly detect shifts in sentiment, narrative framing, or socio-cultural implications - aspects that are critical when assessing demographic bias. These metrics should be interpreted as proxies rather than definitive indicators of fairness or bias. Therefore, we supplemented their results with further qualitative analysis such as AIR and sentiment evaluation.

### **Bias from Sequential LLM Processing (Two-Step Risk)**

Our methodology requires running inputs through the LLM twice: first to modify the demographic identity in the story, and then to generate a summary from the modified version. This two-step processing introduces additional risk, as the second summary generation may reflect compounded biases or artifacts introduced during the first step. For example, if the LLM subtly alters tone or structure during the identity change, those differences may carry forward into the summary, magnifying disparities. This layered processing makes it harder to isolate the root cause of any observed bias. Future iterations could explore more direct identity attribute control methods, such as fine-tuning the LLM on targeted demographic edits, or leveraging structured prompt engineering to minimize generation drift.

## **Conclusion**

This project developed a structured and reproducible framework to assess fairness in LLM-generated summaries, with a focus on identifying disparities linked to demographic variations in race and gender. By systematically modifying character identities within curated financial narratives, generating

summaries using the Cohere Command R model on AWS Bedrock, and applying evaluation metrics such as ROUGE, BERTScore, and statistical analysis techniques including MANOVA and ANOVA, we analyzed whether identity shifts influenced the tone or content of model outputs.

Our results suggest that while most demographic modifications did not lead to large-scale disparities, as indicated by average AIR ratios above threshold, localized differences were observed, particularly between white\_male and white\_female summaries in ROUGE-2 F1 scores and sentiment analysis. These findings demonstrate that even subtle shifts in language and framing can occur based on demographic identity, emphasizing the importance of supplementing quantitative metrics with qualitative review.

The playbook presented in this report offers financial institutions a scalable, modular approach for auditing AI-generated outputs for fairness. Future work could enhance the framework through structured red teaming, field testing with real-world inputs, and human-calibrated evaluation processes to further strengthen its reliability and applicability across broader contexts.

## Appendices

### Appendix A - BERT F1 ANOVA Model Results

Model Summary:

| OLS Regression Results                                                  |                  |                     |          |         |       |        |        |
|-------------------------------------------------------------------------|------------------|---------------------|----------|---------|-------|--------|--------|
| Dep. Variable:                                                          | BERT_F1          | R-squared:          | 0.013    |         |       |        |        |
| Model:                                                                  | OLS              | Adj. R-squared:     | 0.005    |         |       |        |        |
| Method:                                                                 | Least Squares    | F-statistic:        | 1.585    |         |       |        |        |
| Date:                                                                   | Tue, 08 Apr 2025 | Prob (F-statistic): | 0.177    |         |       |        |        |
| Time:                                                                   | 23:57:57         | Log-Likelihood:     | 758.79   |         |       |        |        |
| No. Observations:                                                       | 500              | AIC:                | -1508.   |         |       |        |        |
| Df Residuals:                                                           | 495              | BIC:                | -1487.   |         |       |        |        |
| Df Model:                                                               | 4                |                     |          |         |       |        |        |
| Covariance Type:                                                        | nonrobust        |                     |          |         |       |        |        |
|                                                                         |                  | coef                | std err  | t       | P> t  | [0.025 | 0.975] |
| Intercept                                                               |                  | 0.7046              | 0.005    | 132.154 | 0.000 | 0.694  | 0.715  |
| C(protected_class, Treatment(reference='white_male2'))[T.asian_male]    |                  | -0.0036             | 0.008    | -0.483  | 0.630 | -0.018 | 0.011  |
| C(protected_class, Treatment(reference='white_male2'))[T.black_male]    |                  | -0.0115             | 0.008    | -1.521  | 0.129 | -0.026 | 0.003  |
| C(protected_class, Treatment(reference='white_male2'))[T.hispanic_male] |                  | 0.0032              | 0.008    | 0.430   | 0.667 | -0.012 | 0.018  |
| C(protected_class, Treatment(reference='white_male2'))[T.white_female]  |                  | -0.0116             | 0.008    | -1.542  | 0.124 | -0.026 | 0.003  |
| Omnibus:                                                                | 213.763          | Durbin-Watson:      | 1.406    |         |       |        |        |
| Prob(Omnibus):                                                          | 0.000            | Jarque-Bera (JB):   | 2270.162 |         |       |        |        |
| Skew:                                                                   | -1.561           | Prob(JB):           | 0.00     |         |       |        |        |
| Kurtosis:                                                               | 12.961           | Cond. No.           | 5.83     |         |       |        |        |

Note. All p-values are greater than .05 for this dataset showing no statistically significant.

### Appendix B - BERT F1 Scheffe Grouping Letter Results

Scheffé Grouping Letters:  
 Group asian\_male: A B C  
 Group black\_male: B D E  
 Group hispanic\_male: C E F  
 Group white\_female: B D G  
 Group white\_male2: A B C E F H

Note. Groups that share the same Scheffe grouping letter are not significantly different. All groups have overlapping letters in some capacity so this confirms that no protected class is significantly different from another in terms of mean BERT F1 score.

### Appendix C - ROUGE-1 F1 Scheffe Grouping Letters Results

Scheffé Grouping Letters:  
 Group asian\_male: A  
 Group black\_male: A B C  
 Group hispanic\_male: A C D  
 Group white\_female: B D E  
 Group white\_male2: A B C D F

Note. Groups that share the same Scheffe grouping letter are not significantly different. All groups have overlapping letters in some capacity so this confirms that no protected class is significantly different from another in terms of mean ROUGE-1 F1 score. Scheffe grouping letters are stricter than p-values.

## Appendix D - ROUGE-2 F1 ANOVA Model Results

Model Summary:

| OLS Regression Results                                                   |                  |                     |          |       |        |        |
|--------------------------------------------------------------------------|------------------|---------------------|----------|-------|--------|--------|
| Dep. Variable:                                                           | ROUGE_2_F1       | R-squared:          | 0.021    |       |        |        |
| Model:                                                                   | OLS              | Adj. R-squared:     | 0.013    |       |        |        |
| Method:                                                                  | Least Squares    | F-statistic:        | 2.698    |       |        |        |
| Date:                                                                    | Tue, 08 Apr 2025 | Prob (F-statistic): | 0.0302   |       |        |        |
| Time:                                                                    | 23:58:56         | Log-Likelihood:     | 510.27   |       |        |        |
| No. Observations:                                                        | 500              | AIC:                | -1011.   |       |        |        |
| Df Residuals:                                                            | 495              | BIC:                | -989.5   |       |        |        |
| Df Model:                                                                | 4                |                     |          |       |        |        |
| Covariance Type:                                                         | nonrobust        |                     |          |       |        |        |
|                                                                          | coef             | std err             | t        | P> t  | [0.025 | 0.975] |
| Intercept                                                                | 0.1466           | 0.009               | 16.731   | 0.000 | 0.129  | 0.164  |
| C(protected_class, Treatment(reference='white_male2')) [T.asian_male]    | 0.0099           | 0.012               | 0.803    | 0.423 | -0.014 | 0.034  |
| C(protected_class, Treatment(reference='white_male2')) [T.black_male]    | 0.0018           | 0.012               | 0.141    | 0.888 | -0.023 | 0.026  |
| C(protected_class, Treatment(reference='white_male2')) [T.hispanic_male] | 0.0213           | 0.012               | 1.716    | 0.087 | -0.003 | 0.046  |
| C(protected_class, Treatment(reference='white_male2')) [T.white_female]  | -0.0179          | 0.012               | -1.444   | 0.149 | -0.042 | 0.006  |
| Omnibus:                                                                 | 64.734           | Durbin-Watson:      | 1.389    |       |        |        |
| Prob(Omnibus):                                                           | 0.000            | Jarque-Bera (JB):   | 95.407   |       |        |        |
| Skew:                                                                    | 0.870            | Prob(JB):           | 1.92e-21 |       |        |        |
| Kurtosis:                                                                | 4.246            | Cond. No.           | 5.83     |       |        |        |

Note. All p-values are greater than .05 for this dataset showing no statistically significant.

## Appendix E - ROUGE-2 F1 Scheffe Grouping Letter Results

Scheffé Grouping Letters:  
 Group asian\_male: A  
 Group black\_male: A B C  
 Group hispanic\_male: A C D E  
 Group white\_female: B D E F  
 Group white\_male2: A B C F G

Note. Groups that share the same Scheffe grouping letter are not significantly different. All groups have overlapping letters in some capacity so this confirms that no protected class is significantly different from another in terms of mean ROUGE-2 F1 score.



## Appendix F - ROUGE-L F1 ANOVA Model Results

Model Summary:

| OLS Regression Results                                                   |                  |                     |          |       |        |        |
|--------------------------------------------------------------------------|------------------|---------------------|----------|-------|--------|--------|
| Dep. Variable:                                                           | ROUGE_L_F1       | R-squared:          | 0.018    |       |        |        |
| Model:                                                                   | OLS              | Adj. R-squared:     | 0.010    |       |        |        |
| Method:                                                                  | Least Squares    | F-statistic:        | 2.276    |       |        |        |
| Date:                                                                    | Tue, 08 Apr 2025 | Prob (F-statistic): | 0.0601   |       |        |        |
| Time:                                                                    | 23:59:18         | Log-Likelihood:     | 512.61   |       |        |        |
| No. Observations:                                                        | 500              | AIC:                | -1015.   |       |        |        |
| Df Residuals:                                                            | 495              | BIC:                | -994.1   |       |        |        |
| Df Model:                                                                | 4                |                     |          |       |        |        |
| Covariance Type:                                                         | nonrobust        |                     |          |       |        |        |
|                                                                          | coef             | std err             | t        | P> t  | [0.025 | 0.975] |
| Intercept                                                                | 0.2624           | 0.009               | 30.080   | 0.000 | 0.245  | 0.280  |
| C(protected_class, Treatment(reference='white_male2')) [T.asian_male]    | 0.0109           | 0.012               | 0.883    | 0.378 | -0.013 | 0.035  |
| C(protected_class, Treatment(reference='white_male2')) [T.black_male]    | -0.0016          | 0.012               | -0.128   | 0.898 | -0.026 | 0.023  |
| C(protected_class, Treatment(reference='white_male2')) [T.hispanic_male] | 0.0191           | 0.012               | 1.546    | 0.123 | -0.005 | 0.043  |
| C(protected_class, Treatment(reference='white_male2')) [T.white_female]  | -0.0155          | 0.012               | -1.258   | 0.209 | -0.040 | 0.009  |
| Omnibus:                                                                 | 61.140           | Durbin-Watson:      | 1.343    |       |        |        |
| Prob(Omnibus):                                                           | 0.000            | Jarque-Bera (JB):   | 102.084  |       |        |        |
| Skew:                                                                    | 0.769            | Prob(JB):           | 6.80e-23 |       |        |        |
| Kurtosis:                                                                | 4.592            | Cond. No.           | 5.83     |       |        |        |

*Note. All p-values are greater than .05 for this dataset showing no statistically significant.*

## Appendix G - ROUGE-L F1 Scheffe Grouping Letter Results

Scheffé Grouping Letters:

Group asian\_male: A B

Group black\_male: B C D

Group hispanic\_male: A D E F

Group white\_female: C E F G

Group white\_male2: A B C D G H

*Note. Groups that share the same Scheffe grouping letter are not significantly different. All groups have overlapping letters in some capacity so this confirms that no protected class is significantly different from another in terms of mean ROUGE-L F1 score.*

## Appendix H – AIR Results

| Protected Class | Score Type     | Avg AIR Ratio | Flag            |
|-----------------|----------------|---------------|-----------------|
| asian_male      | BERT_F1        | 0.998         | Above Threshold |
| asian_male      | BERT_Precision | 0.998         | Above Threshold |
| asian_male      | BERT_Recall    | 0.9985        | Above Threshold |
| asian_male      | ROUGE_1_F1     | 1.0356        | Above Threshold |
| asian_male      | ROUGE_2_F1     | 1.5211        | Above Threshold |

|               |                |        |                 |
|---------------|----------------|--------|-----------------|
| asian_male    | ROUGE_L_F1     | 1.0968 | Above Threshold |
| black_male    | BERT_F1        | 0.9865 | Above Threshold |
| black_male    | BERT_Precision | 0.9899 | Above Threshold |
| black_male    | BERT_Recall    | 0.9842 | Above Threshold |
| black_male    | ROUGE_1_F1     | 1.0111 | Above Threshold |
| black_male    | ROUGE_2_F1     | 1.4375 | Above Threshold |
| black_male    | ROUGE_L_F1     | 1.0545 | Above Threshold |
| hispanic_male | BERT_F1        | 1.0074 | Above Threshold |
| hispanic_male | BERT_Precision | 1.0078 | Above Threshold |
| hispanic_male | BERT_Recall    | 1.0075 | Above Threshold |
| hispanic_male | ROUGE_1_F1     | 1.0555 | Above Threshold |
| hispanic_male | ROUGE_2_F1     | 1.515  | Above Threshold |
| hispanic_male | ROUGE_L_F1     | 1.1275 | Above Threshold |
| white_female  | BERT_F1        | 0.9867 | Above Threshold |
| white_female  | BERT_Precision | 0.9871 | Above Threshold |
| white_female  | BERT_Recall    | 0.9873 | Above Threshold |
| white_female  | ROUGE_1_F1     | 0.9628 | Above Threshold |
| white_female  | ROUGE_2_F1     | 1.2422 | Above Threshold |
| white_female  | ROUGE_L_F1     | 1.0044 | Above Threshold |
| white_male2   | BERT_F1        | 1      | Above Threshold |
| white_male2   | BERT_Precision | 1      | Above Threshold |
| white_male2   | BERT_Recall    | 1      | Above Threshold |
| white_male2   | ROUGE_1_F1     | 1      | Above Threshold |
| white_male2   | ROUGE_2_F1     | 1      | Above Threshold |
| white_male2   | ROUGE_L_F1     | 1      | Above Threshold |

*Note. The table shown average AIR results by protected class and score type. The far right column states whether the score is above or below the .9 threshold.*

**Appendix I - Top 10 Most Negative and Positive Sentiment Score Differences between white\_male1 and white\_female**

| <b>Top 10 Most Negative Sentiment Score Summary Differences (white_male1 and white_female)</b> | <b>Sentiment</b> |
|------------------------------------------------------------------------------------------------|------------------|
| A man claiming to be basketball star James Joh...                                              | -0.3             |
| Staff Sgt. James Johnson has been charged with...                                              | -0.5             |
| Russian oligarchs who've opposed the Kremlin h...                                              | -0.26            |
| Internet millionaire James Johnson, known for ...                                              | -0.0568          |
| Two men have been given record sentences for t...                                              | -0.0778          |
| Staff Sgt. James Johnson, accused of killing A...                                              | -0.25            |
| Staff Sgt. James Johnson was shot by a sniper ...                                              | 0.0189           |
| A man from Augusta, Georgia, known as 'Kilobit...                                              | -0.1611          |
| James Johnson, Britain's Bill Gates, denies ac...                                              | -0.2167          |
| A prominent member of Anonymous and LulzSec, J...                                              | -0.13            |
| <b>Top 10 Most Positive Sentiment Score Summary Differences (white_male1 and white_female)</b> | <b>Sentiment</b> |
| The model being described is Command_large, a ...                                              | 0.1905           |
| Federal prosecutors are requesting a Florida j...                                              | 0.4167           |
| James Johnson and his wife Michelle have been ...                                              | 0.16             |
| Sure! Here's a rewritten version of the transc...                                              | 0.2969           |
| The IRS has awarded a record-breaking \$104 mil...                                             | 0.175            |
| A global health advocate, James Johnson, witne...                                              | 0.0111           |
| A father-son duo, James and Michael Johnson, r...                                              | -0.0208          |
| Some of history's most notorious figures includ...                                             | 0.1212           |
| Three successful professionals were caught for...                                              | 0.3              |
| James Johnson lived a double life, pretending ...                                              | 0.3              |

*Note. The top 10 cases with the most negative sentiment score indicate a lower sentiment score for white\_male1 compared to white\_female for the corresponding summary. The second list of top 10 cases with the most positive sentiment score indicate higher sentiment scores for white\_male 1 compared to white\_female for the corresponding summary.*

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